

Regular paper

A 0.92 ppm/°C, 1-V supply curvature-compensated bandgap reference using mutual compensation of V_{GS} and V_{BE} as dual CTAT components

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ABSTRACT

This paper presents a curvature-compensated bandgap reference (BGR) that achieves a 0.92-ppm/°C temperature coefficient (TC) through a modified design approach derived from the conventional Banba BGR architecture. The proposed topology introduces an additional complementary to absolute-temperature (CTAT) component derived from the MOSFET gate-source voltage (V_{GS}) biased in the subthreshold region, in addition to the conventional CTAT component generated by V_{BE} . The combination of these two CTAT components effectively compensates the nonlinear temperature dependence of V_{BE} . By properly scaling a proportional to absolute temperature (PTAT) component extracted from ΔV_{BE} together with the CTAT components derived from V_{BE} and V_{GS} , the overall temperature dependence of the reference voltage is significantly reduced, resulting in a near-zero TC. The proposed structure also enables independent trimming of the output voltage level and the TC. Post-layout PVT simulations and a 500-run Monte Carlo analysis demonstrate an average reference voltage of approximately 600 mV and an average TC of 0.92 ppm/°C. The proposed BGR further achieves a 0.003%/V LS over a 1–2 V supply with a power consumption of 9.1 μ W.

1. Introduction

Voltage references are key circuits that provide stability and accuracy in modern electronic systems. They are widely employed in high-resolution analog-to-digital converters (ADCs), power management circuits, temperature and environmental sensors, precision battery monitoring systems, and numerous other integrated and mixed-signal circuits [1–4]. Their stability directly impacts the overall accuracy and reliability of these systems. Depending on the target application, different techniques can be employed to realize a voltage reference block. In ultra-low-power and low-voltage systems, bipolar transistors (BJTs) are generally avoided, and MOS-only architectures are preferred because of their better compatibility with advanced CMOS technologies. However, purely MOS-based references usually exhibit relatively high temperature coefficients (TCs), often on the order of tens of ppm/°C [5,6]. This is because the PTAT and CTAT components are typically generated from subthreshold MOS gate-source voltage differences and the gate-source voltage itself, respectively [7,8], making accurate temperature compensation difficult even when curvature-compensation techniques are employed [9,10]. Although accurate MOS-only references based on zero-temperature-coefficient (ZTC) MOSFET operation have also been reported, these circuits typically consume power in the

microwatt range and often lack output-voltage trimming capability [11,12].

On the other hand, several voltage references with very low TCs have been reported based on curvature-compensated bandgap voltage references (BGRs) implemented using BJTs. Conventional BGR relies on ΔV_{BE} as a PTAT term, which is approximately linear with temperature, and V_{BE} as a CTAT term with a nonlinear temperature dependence. However, conventional BGRs typically require a supply voltage higher than 1.5 V and achieve TCs around 10 ppm/°C. The Banba BGR proposed in [13] enables operation from supply voltages close to 1 V while maintaining comparable TC performance. To further reduce TC, various techniques have been proposed. In [14,15], attempts have been made to compensate and linearize the nonlinear CTAT component of V_{BE} ; however, both approaches require supply voltages of around 3 V. Several additional compensation techniques have also been reported in [16–18], but they are either unsuitable for low-supply operation or, as in [19], require a large active area and often involve complex circuit topologies. Moreover, most reported sub-ppm/°C voltage references rely on multi-stage compensation schemes and typically consume tens of microamperes of current. In many of these techniques, the bias current also increases with temperature due to the temperature dependence of V_{BE} related biasing mechanisms [17,19]. Achieving simultaneously low supply voltage, low

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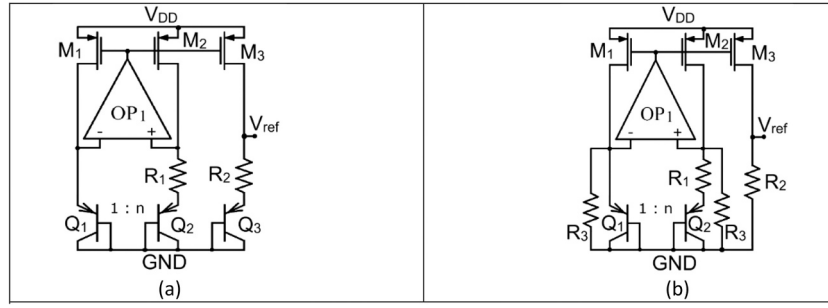


Fig. 1. (a) Conventional bandgap reference (BGR), (b) Banba bandgap reference topology.

power consumption, compact implementation, and high thermal stability therefore remains a significant challenge in BGR design.

This paper presents an ultra-low-TC bandgap reference derived from the conventional Banba BGR, supporting operation down to a 1-V supply. Unlike previous low-voltage hybrid BGRs, the proposed approach exploits the opposite nonlinear temperature characteristics of V_{BE} and V_{GS} to realize intrinsic curvature compensation within a 1 V Banba-type architecture. It is shown that replacing V_{BE} with V_{GS} in a Banba BGR results in a concave temperature characteristic, whereas the conventional Banba structure exhibits a convex temperature behavior. By simultaneously utilizing both V_{GS} and V_{BE} as CTAT components, significant curvature reduction and improved thermal stability are achieved. In addition, the proposed BGR provides independent trimming capability for both the output voltage and TC, while output-voltage trimming has minimal impact on thermal performance. Owing to its relatively simple architecture, the proposed circuit also achieves improved power efficiency compared to many recently reported sub-ppm references.

This paper is organized as follows. Section 2 briefly reviews the Banba BGR and introduces the basic concepts. Section 3 describes the proposed BGR in detail, and Section 4 presents the simulation results.

2. Banba BGR review and fundamental concept of the proposed circuit

This section first reviews the Banba BGR circuit and examines the behavior of its CTAT component (V_{BE}), which exhibits a nonlinear dependence on temperature. Next, a Banba-type BGR whose CTAT term is generated using V_{GS} is investigated. Finally, the basic concept and a preliminary implementation of the proposed approach are introduced, where a combination of V_{GS} and V_{BE} is employed as the CTAT component. While Section 4 presents post-layout results of the final circuit, schematic-level (pre-layout) simulations are used here to illustrate the proposed concepts and support the analysis.

2.1. Brief review of the Banba BGR

Fig. 1(a) shows a conventional 1.2 V BGR requiring about 1.5 V supply [20]. The Banba BGR in Fig. 1(b) reduces the minimum supply voltage to approximately 1 V [13].

Assuming that the emitter area ratio of Q_2 to Q_1 is equal to n , and that transistors M_1 , M_2 , and M_3 are identical in Fig. 1(b), the reference voltage of the Banba BGR can be expressed as

$$V_{ref} = R_2 I_{D3} = R_2 \left(\frac{V_T \ln(n)}{R_1} + \frac{V_{EB1}}{R_3} \right) \quad (1)$$

where V_T denotes the thermal voltage (kT/q), where k is the Boltzmann constant, T is the absolute temperature, and q is the electron charge. A similar expression can also be derived for the conventional BGR shown in Fig. 1(a) as

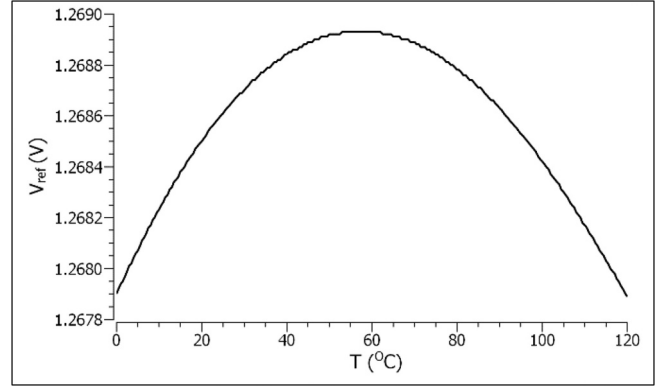


Fig. 2. Simulated output voltage versus temperature for the circuit of Fig. 1(a) with $n = 8$, $R_1 = 50$ k Ω , and $R_2 = 581$ k Ω .

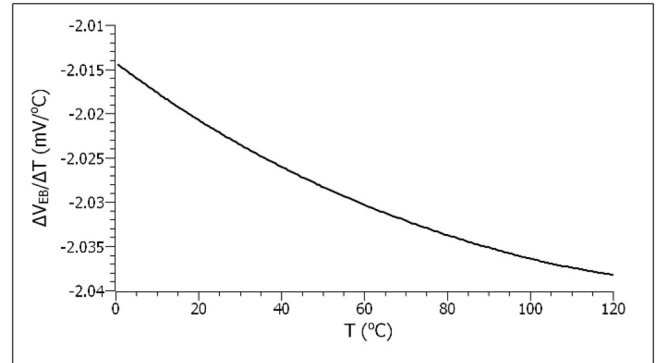


Fig. 3. Simulated $\Delta V_{EB3}/\Delta T$ versus temperature.

$$V_{ref} = R_2 I_{D3} + V_{EB3} = \frac{R_2}{R_1} V_T \ln(n) + V_{EB3} \quad (2)$$

From Eq. (1) and Eq. (2), adjusting R_2 in the Banba BGR enables low-voltage operation. Start-up circuits are omitted for simplicity. As shown in detail in Section 3.5.1, the CTAT component (V_{BE}) is a nonlinear function of temperature, such that $|dV_{BE}/dT|$ increases with temperature. Hence, assuming the zero-TC point is set at T_0 (approximately between T_{min} and T_{max}), since the PTAT term is linear and the CTAT slope increases with temperature, the output becomes convex, reaching its maximum at T_0 and decreasing toward both T_{min} and T_{max} .

For better illustration, the circuit of Fig. 1(a) is simulated with $n = 8$, $R_1 = 50$ k Ω , and $R_2 = 581$ k Ω , using an op-amp that will be described in detail in Section 3, with $V_{DD} = 1.5$ V. The result is shown in Fig. 2. The output voltage is 1.27 V and the obtained TC is about 7 ppm/°C. In addition, Fig. 3 also shows $\Delta V_{EB3}/\Delta T$. As can be seen, $\Delta V_{EB3}/\Delta T$ changes

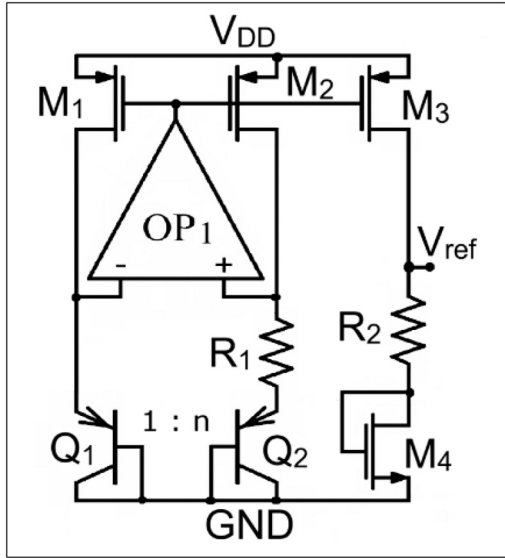


Fig. 4. BGR circuit employing V_{GS} as the CTAT component.

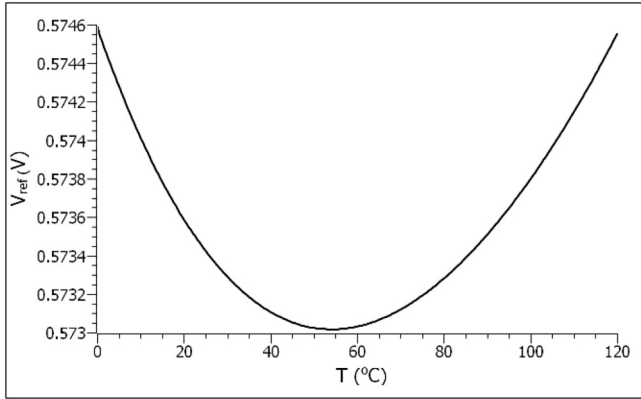


Fig. 5. Simulated output reference voltage of the BGR shown in Fig. 4.

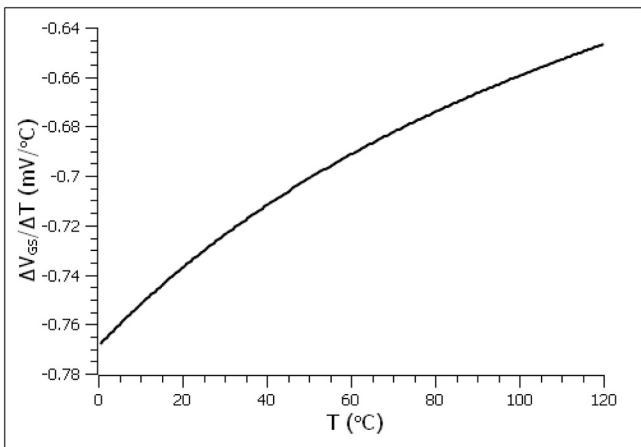


Fig. 6. Temperature variation of $\Delta V_{GS}/\Delta T$.

from $-2.01 \text{ mV}/^\circ\text{C}$ to $-2.04 \text{ mV}/^\circ\text{C}$ over a 100°C temperature range, confirming the previous discussion.

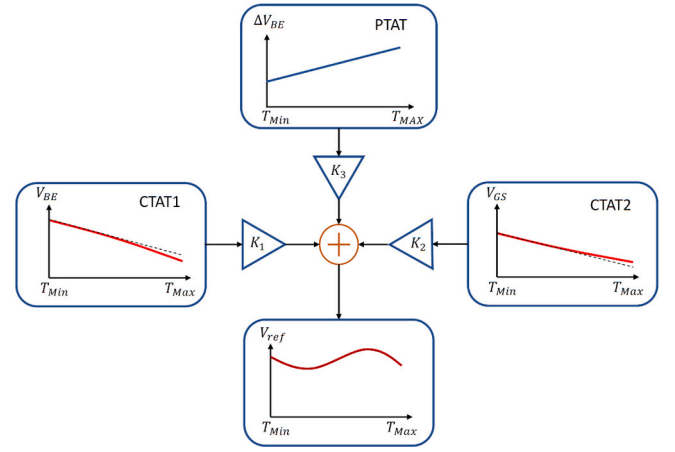


Fig. 7. Conceptual curvature compensation using complementary nonlinearities of V_{BE} , V_{GS} .

2.2. Banba-type BGR using V_{GS} as the CTAT term

The bandgap reference shown in Fig. 4 is similar to the conventional topology, except that the CTAT component, which is V_{BE} in the conventional circuit, is replaced by the V_{GS4} of a diode-connected nMOS transistor (M_4) biased in the subthreshold region. As shown in detail in Section 3.5.2, the CTAT component (V_{GS}) is a nonlinear function of temperature, such that $|dV_{GS}/dT|$ decreases with temperature. Consequently, the BGR in Fig. 4 exhibits a concave temperature characteristic, unlike a conventional BGR.

For clearer illustration, the circuit shown in Fig. 4 is simulated with $n = 8$, $R_1 = 50 \text{ k}\Omega$, $R_2 = 199 \text{ k}\Omega$, $W_4 = 20 \mu\text{m}$, $L_4 = 1 \mu\text{m}$, and $V_{DD} = 1 \text{ V}$. The simulation results are presented in Fig. 5. The generated reference voltage is 594 mV, and the resulting TC is approximately $23 \text{ ppm}/^\circ\text{C}$. Although the TC is higher than that of the conventional BGR, the minimum supply voltage is reduced to below 1 V. Furthermore, Fig. 6 shows the variation of $\Delta V_{GS}/\Delta T$. It can be observed that $\Delta V_{GS}/\Delta T$ changes from $-0.75 \text{ mV}/^\circ\text{C}$ to $-0.65 \text{ mV}/^\circ\text{C}$ over a 100°C temperature range, confirming the previous discussion. As can be seen, $\Delta V_{GS4}/\Delta T$ exhibits an increasing trend with temperature, whereas $\Delta V_{BE3}/\Delta T$ shows a decreasing trend, as expected.

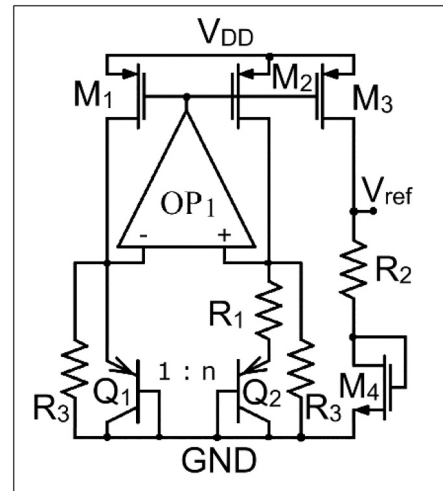


Fig. 8. Preliminary curvature-compensated BGR using V_{BE} - and V_{GS} -based CTAT terms.

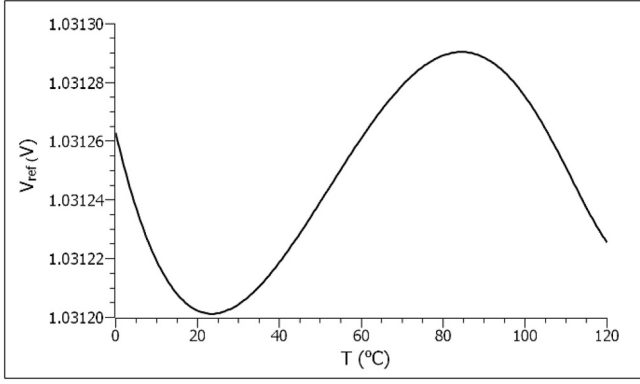


Fig. 9. Output voltage of the preliminary curvature-compensated circuit (TC = 0.7 ppm/°C).

2.3. Preliminary curvature compensation using combined VBE and VGS CTAT terms

As discussed earlier, V_{BE} and V_{GS} exhibit different nonlinear temperature behaviors. The magnitude of the negative slope of V_{BE} slightly increases with temperature, whereas the temperature dependence of V_{GS} shows the opposite trend. This complementary behavior suggests that an appropriate combination of these two CTAT quantities can partially compensate for the curvature of the reference voltage.

Fig. 7 conceptually illustrates the temperature characteristics of the involved components. One CTAT term is proportional to V_{BE} , while the second CTAT component is generated from V_{GS} . A PTAT term proportional to the thermal voltage V_T is also generated from ΔV_{BE} , similar to the Banba BGR. By properly scaling these three components, their temperature dependencies can be combined to shape the overall temperature behavior of the reference voltage, resulting in minimal temperature dependence.

Based on this concept, a preliminary curvature-compensated version of the Banba BGR is presented in Fig. 8, where the output voltage V_{ref} can be written as follows:

$$V_{ref} = R_2 I_{D3} + V_{GS4} = \frac{R_2}{R_1} V_T \ln(n) + \frac{R_2}{R_3} V_{EB1} + V_{GS4} \quad (3)$$

To verify this concept, the circuit was simulated with $R_1 = 50 \text{ k}\Omega$, $R_2 = 448.3 \text{ k}\Omega$, $R_3 = 960 \text{ k}\Omega$, and an nMOS transistor M_4 with $W_4 = 10 \text{ }\mu\text{m}$ and $L_4 = 1 \text{ }\mu\text{m}$ using a low- V_{TH} device to reduce the minimum supply voltage. The simulated result shown in Fig. 9 yields $V_{ref} = 1.03 \text{ V}$ with an excellent TC of 0.7 ppm/°C. Although this preliminary structure demonstrates effective curvature compensation, it still requires a relatively high supply voltage and lacks a voltage trimming circuit. The following section introduces an improved topology that resolves these limitations.

3. Proposed curvature-compensated BGR

This section presents the proposed curvature-compensated BGR. First, the circuit architecture and the derivation of the reference voltage are described. Next, the trimming approach for both the TC and output level is discussed. Then, the start-up circuit and the power supply ripple rejection (PSRR) performance are analyzed. Finally, the nonlinear temperature behavior of V_{GS} and V_{BE} is discussed, and their complementary curvature is explained.

3.1. Proposed BGR and output derivation

Fig. 10(a) shows the proposed BGR circuit, which is an improved version of the previous design in Fig. 8. The new configuration operates from a reduced supply voltage of 1 V and incorporates trimming circuits for both the output voltage and the TC. The TC-trimming circuit and the output-voltage-trimming array are shown in Fig. 10(b) and (c), respectively.

As can be seen, pseudo-cascode transistors are employed in the current-source branches to increase the output resistance and thereby reduce the dependence of the reference voltage on the supply level and its variations. If transistors M_{1a} , M_{2a} , M_{3a} , and M_{4a} are matched, and low-threshold MOS transistors M_{1b} , M_{2b} , M_{3b} , and M_{4b} are also matched, the drain current of all eight devices becomes equal to $(V_T \ln(n)/R_1 + V_{EB1}/R_2)$. Thus, assuming $n = 8$, the reference output voltage can be expressed as

$$\begin{aligned} V_{ref} &= V_{GS5} \frac{R_4}{R_3 + R_4} + I_{D4b} (R_3 \parallel R_4) \\ &= \frac{R_4}{R_3 + R_4} \left[V_{GS5} + R_3 \left(\frac{V_{EB1}}{R_2} + \frac{V_T \ln(8)}{R_1} \right) \right]. \end{aligned} \quad (4)$$

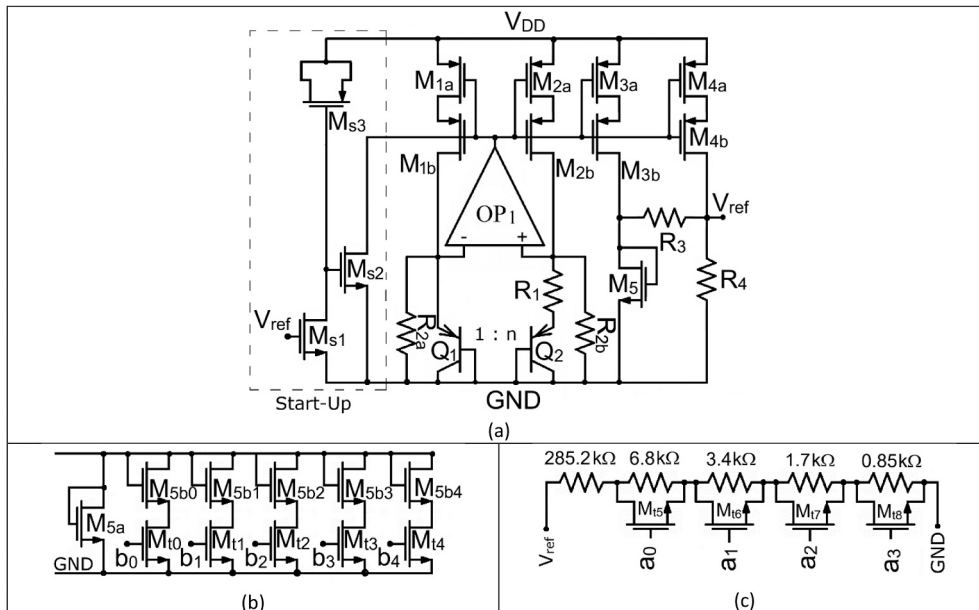


Fig. 10. (a) Proposed curvature-compensated BGR, (b) TC-trimming circuit using segmented transistor M_5 , (c) Output-voltage trimming using resistor array R_4 .

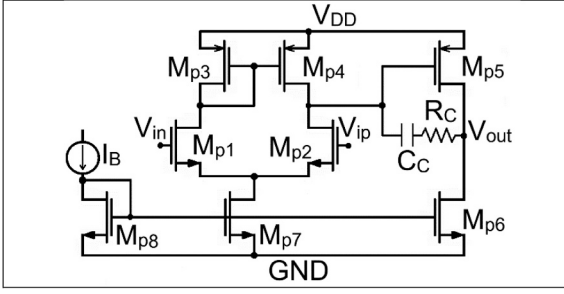


Fig. 11. Two stage OpAmp circuit.

Table 1
Parameters of key circuit components.

Component	Parameter	Component	Parameter
$M_{1a}, M_{2a}, M_{3a}, M_{4a}$	60 $\mu\text{m}/1 \mu\text{m}$, $F = 5$	M_{p3}, M_{p4}	80 $\mu\text{m}/5 \mu\text{m}$, $F = 2$
$M_{1b}, M_{2b}, M_{3b}, M_{4b}$ (Low V_{TH})	24 $\mu\text{m}/2 \mu\text{m}$, $F = 2$	M_{p5}	160 $\mu\text{m}/5 \mu\text{m}$, $F = 4$
M_{s1}	0.5 $\mu\text{m}/5 \mu\text{m}$, $F = 1$	M_{p6}, M_{p7}	40 $\mu\text{m}/2 \mu\text{m}$, $F = 2$
M_{s2}	0.3 $\mu\text{m}/20 \mu\text{m}$, $F = 1$	M_{p8}	20 $\mu\text{m}/2 \mu\text{m}$, $F = 1$
M_{s3}	10 $\mu\text{m}/10 \mu\text{m}$, $F = 1$	R_1	45.85 k Ω
M_{sa}	9.8 $\mu\text{m}/1 \mu\text{m}$, $F = 1$	R_2	627.1 k Ω
M_{sb0}	16 $\mu\text{m}/1 \mu\text{m}$, $F = 1$	R_3	1090 k Ω
M_{sb1}	8 $\mu\text{m}/1 \mu\text{m}$, $F = 1$	R_4	Reported in Fig. 10(c)
M_{sb2}	4 $\mu\text{m}/1 \mu\text{m}$, $F = 1$	R_C	6 k Ω
M_{sb3}	2 $\mu\text{m}/1 \mu\text{m}$, $F = 1$	C_C	0.95 pF
M_{sb4}	1 $\mu\text{m}/1 \mu\text{m}$, $F = 1$	I_B	50 nA
$M_{t0}, M_{t1}, \dots, M_{t7}, M_{t8}$	10 $\mu\text{m}/0.25 \mu\text{m}$, $F = 1$	Q_1	$10 \times 10 \mu\text{m}^2$
M_{p1}, M_{p2}	120 $\mu\text{m}/5 \mu\text{m}$, $F = 3$	Q_2	$8 \times (10 \times 10 \mu\text{m}^2)$

By properly adjusting R_2 and R_3 , the desired curvature-compensated characteristic previously shown in Fig. 7 is obtained. The operational amplifier circuit shown in Fig. 11 is a two-stage OTA biased in the subthreshold region with a bias current $I_B = 50$ nA. All transistor dimensions and design parameters are summarized in Table 1.

3.2. Trimming

As can be observed in Eq. (4), TC trimming can be achieved by adjusting the size of M_5 . As shown in Fig. 10(b), transistor M_5 consists of six-unit transistors connected in parallel and controlled by a 5-bit digital

TC-trimming word $b = b_0b_1b_2b_3b_4$, which provides 32 selectable states. As we will see in Section 3.5.2, the TC of a subthreshold-biased V_{GS} can be adjusted by changing the W/L ratio. The proposed TC-trimming circuit is based on this principle.

To adjust the output level, resistor R_4 is used for output-voltage trimming. As indicated in Eq. (4), R_4 affects the output voltage but does not influence the TC; therefore, it can be used to adjust the output voltage independently. This enables a two-step trimming procedure in which the TC is first adjusted using M_5 , followed by fine-tuning of the output voltage through R_4 . The output-voltage trimming network (R_4 array) is shown in Fig. 10(c). As illustrated, a 4-bit digital output-magnitude control word $a = a_0a_1a_2a_3$, provides 16 discrete trimming steps with a resolution of 0.85 k Ω , allowing R_4 to vary from 285.2 k Ω to 310.87 k Ω .

It should be noted that changing the value of R_4 during trimming slightly modifies the current flowing through M_5 , which may in principle affect the TC. However, this effect is negligible because resistor R_3 (approximately 1 M Ω) is much larger than the dynamic resistance of M_5 . Since M_5 operates in the subthreshold region, its dynamic resistance can be approximated by $r_d = \zeta V_T / I_D$. In this design, r_d is approximately 30 k Ω , confirming that the influence of R_4 variation on the TC is minimal.

For TC trimming, all 32 codes of digital word b are swept to determine the optimum and to verify sub-1 ppm/ $^{\circ}\text{C}$ performance. In production, the optimal codes would be permanently stored after a few calibration sweeps, potentially aided by machine-learning-based code selection (e.g., pattern recognition). Then, at room temperature, the output voltage amplitude is adjusted using the voltage trimming code (a).

3.3. Start-up circuit

In Fig. 10(a), the start-up circuit consists of transistors M_{s1} , M_{s2} , and M_{s3} . At power-up, M_{s3} ($W = L = 10 \mu\text{m}$) acts as a capacitor and keeps the gate of M_{s2} near V_{DD} . Initially, M_{s2} keeps the output of the OTA near ground until V_{ref} rises. At that point, M_{s1} turns on and pulls the gate of M_{s2} close to ground, disabling the start-up circuit during normal operation. Another important feature is the OTA structure. As shown in Fig. 11, M_{p6} is the first transistor activated directly by the bias current I_B , forcing the OTA output toward ground during start-up. As the remaining transistors gradually turn on, the OTA reaches its steady-state operating point. Consequently, the combined action of the OTA structure and the start-up circuit maintains the OTA output near ground during start-up, enabling the current mirrors and subsequently turning on BJTs Q_1 and Q_2 .

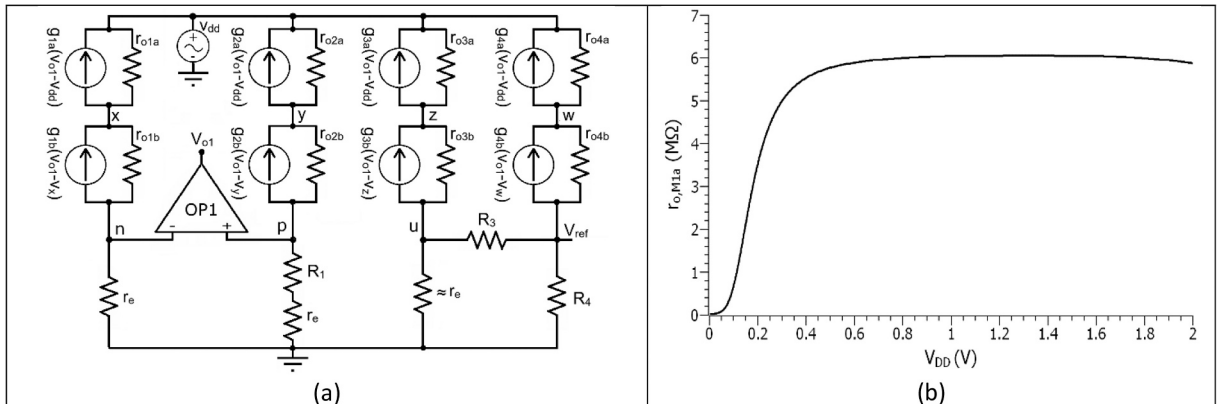


Fig. 12. (a) Simplified small-signal model used for analyzing the supply sensitivity of the reference. (b) Simulated output resistance r_o versus V_{DS} of transistor M_{1a} .

3.4. PSRR and PTAT replication improvement using the Pseudo-Cascode configuration

In this section, the PSRR value and the accuracy of current mirror replication and its importance for achieving a low-TC BGR are analyzed. Only simplified relations are used here to provide insight into the circuit behavior, and a more sophisticated discussion is omitted. The analysis focuses on the effect of V_{DD} on the accuracy of PTAT current replication in the output branch and its role in achieving an ultra-low TC. The small-signal equivalent circuit is shown in Fig. 12(a). It can be shown that.

$R_1 = \ln(n) \times r_{e1} \approx 2.1 \times r_{e1}$, and $r_{e1} = r_{e2}$. Because r_{e1} is much smaller than R_2 , the effect of R_2 is neglected in Fig. 12(a). All MOS transistors operate in the subthreshold region, and the slope factor is assumed to be unity. Therefore, $g_{1a} \approx g_{2a} \approx g_{3a} \approx g_{4a} \approx g_{1b} \approx g_{2b} \approx g_{3b} \approx g_{4b} \approx g_1$.

$r_{o1a} \approx r_{o2a} \approx r_{o3a} \approx r_{o4a} \approx r_o$. Since the channel length of transistors M_{2b} , M_{3b} , and M_{4b} is twice that of M_{1a} , their output resistance can be approximated as $2r_{o1}$.

The open-loop voltage gain of the operational amplifier is denoted by a_o . The exact transfer function obtained from the small-signal analysis is very long and not suitable for practical interpretation. By assuming $r_e < R_2, R_3, r_o$, the PSRR can be approximated by the following simplified expression:

$$PSRR = \frac{V_{ref}}{V_{dd}} \cong 0.476 \frac{R_3 R_4}{a_o r_e (R_3 + R_4 + r_e)} \approx 0.476 \frac{R_3 \parallel R_4}{a_o r_e} \quad (5)$$

A potentially misleading point is that when the PSRR is re-derived for the circuit without the pseudo-cascode current mirrors, a nearly identical expression is obtained. This might suggest that the pseudo-cascode configuration provides little benefit. However, the more critical issue is the accuracy of current-mirror replication in the output branch. Any variation of the output PTAT-CTAT₁ current with supply voltage directly degrades the TC.

At low supply voltages, the source-drain voltages of M_{1b} , M_{2b} , and M_{4b} become very small. Under this condition, their output resistances decrease significantly, which can degrade the TC because the proper weighting of the PTAT, CTAT₁ (V_{EB}/R_2), and CTAT₂ (V_{GS5}) components is disturbed. A detailed analytical treatment of TC sensitivity with respect to V_{DD} is rather involved and beyond the scope of this paper. As an illustration, the nonlinear behavior of the output resistance of a pMOS transistor with the same W/L as M_{1a} and biased at the same V_{GS} used in this design is shown in Fig. 12(b), obtained using a 180-nm standard CMOS technology. As shown, even for V_{DS} values ranging from 0.1 V to 0.4 V, the output resistance r_o continues to vary with V_{DS} due to short-channel effects; for instance, r_o is 684 k Ω at $V_{DS} = 0.1$ V, 3.6 M Ω at $V_{DS} = 0.2$ V, and 6.04 M Ω at $V_{DS} = 1$ V. Accordingly, the accuracy of replication in a simple current mirror is unreliable when V_{DS} is not sufficiently large. In contrast, in a cascode configuration, the drain-source voltage of the upper (cascode) transistor remains nearly constant, ensuring accurate replication. Due to the nonlinearity of the output resistance relationship, we limit our discussion to this extent and defer verification of the current replication accuracy to the simulation results.

3.5. Temperature coefficient analysis of V_{BE} and V_{GS}

Since the nonlinear temperature characteristics of V_{BE} and subthreshold V_{GS} are central to the proposed compensation mechanism, they are discussed in the following two subsections. In both cases, the transistor current is considered proportional to T^α .

3.5.1. Temperature coefficient of V_{BE}

As described in [20,22], the value of V_{BE} when it carries a current proportional to T^α is expressed by the following equation:

$$V_{BE}(T) = V_{G0} - [V_{G0} - V_{BE}(T_r)] \frac{T}{T_r} - (\eta - \alpha) V_T \ln \frac{T}{T_r} \quad (6)$$

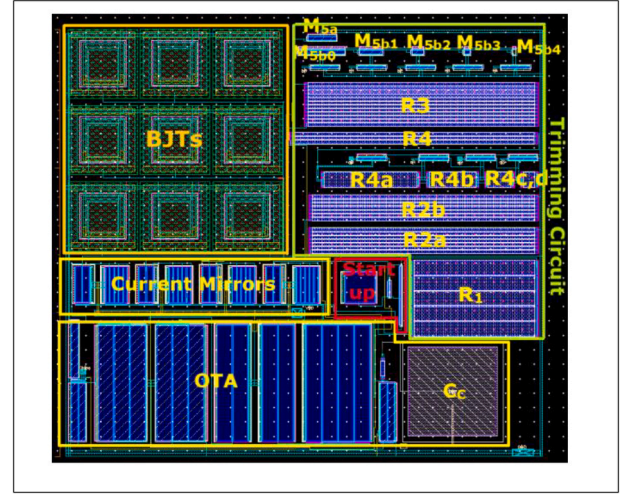


Fig. 13. Layout of the proposed BGR.

where T_r is the reference temperature, V_{G0} is the bandgap voltage of silicon extrapolated to 0° K (approximately 1.2 V), η is a process-dependent constant around 4, α is the exponent of the bias current, and V_T is the thermal voltage. The first two terms represent the linear approximation while the last term (with $\ln(T/T_r)$) accounts for the nonlinear curvature of V_{BE} . Differentiating with respect to T , the temperature coefficient is given by:

$$\frac{\Delta V_{BE}}{\Delta T} = -\frac{[V_{G0} - V_{BE}(T_r)]}{T_r} - \frac{k}{q}(\eta - \alpha) \left(1 + \ln \frac{T}{T_r}\right) \quad (7)$$

This expression shows that $|dV_{BE}/dT|$ becomes larger at higher temperatures due to the nonlinear $\ln(T/T_r)$ term.

3.5.2. Temperature coefficient of V_{GS}

The gate-source voltage of an nMOS transistor operating in the subthreshold region, with V_{DS} greater than 0.1 V, can be expressed by the following Eq. [21].

$$V_{GS} = V_{TH} + \zeta \ln \frac{I_D}{I_0}, I_0 = (\zeta - 1) \mu C_{OX} \frac{W}{L} V_T^2 \quad (8)$$

where V_{TH} is the threshold voltage of the nMOS transistor, ζ is the subthreshold slope factor, μ is the electron mobility, C_{OX} is the gate-oxide capacitance per unit area, W and L are the width and length of the transistor, respectively, and I_D is the drain current. Assuming that the drain current is proportional to T^α and the mobility is also proportional to $T^{(-m)}$, the above relation can be simplified to the following form, where A is a constant coefficient.

$$V_{GS} = V_{TH} + \zeta V_T \ln(A T^{\alpha+m-2}) = V_{TH} + \zeta V_T [\ln A + (\alpha + m - 2) \ln T]. \quad (9)$$

By differentiating the last expression, the temperature coefficient of V_{GS} is obtained. Before that, it should be noted that the subthreshold slope factor (ζ) is typically in the range of 1.2 to 2 and has a weak temperature dependence; therefore, it is often considered constant.

$$\frac{\partial V_{GS}}{\partial T} = \frac{\partial V_{TH}}{\partial T} + \zeta \frac{k}{q} [\ln A + (\alpha + m - 2) (1 + \ln T)] \quad (10)$$

In this expression, the dominant term is dV_{TH}/dT , which is negative and around -1 mV/°C. From the above expression, it can be seen that when $(\alpha + m - 2) > 0$, the magnitude of dV_{GS}/dT decreases nonlinearly with increasing temperature as shown previously in Fig. 6.

The drain current of M_5 , I_{D5} , consists of two PTAT components: I_{D3b} (dominated by $V_T \ln(n)/R_1$) and $I_{R3} = (V_{ref} - V_{GS5})/R_3$. V_{ref} is nearly constant with temperature, while V_{GS5} decreases with temperature; therefore, I_{R3} is also PTAT. Simulations show that I_{D5} increases linearly

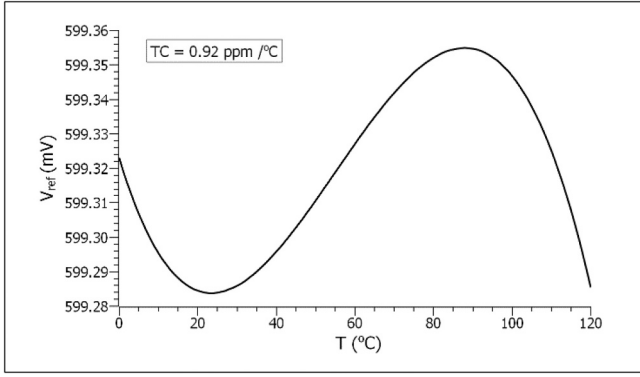


Fig. 14. Simulated reference voltage (V_{ref}) at the TTT process corner using digital trimming words $a = 1001$ and $b = 01000$ at $V_{DD} = 1.1$ V.

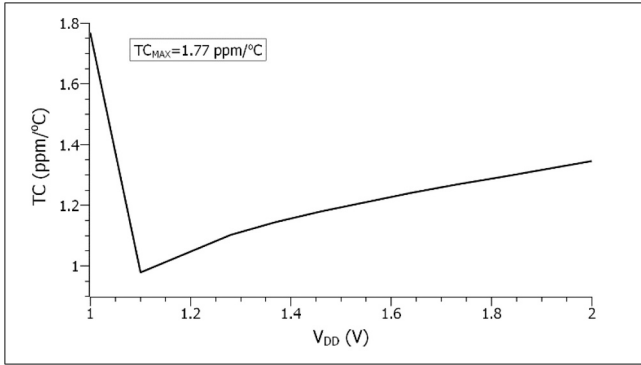


Fig. 15. Temperature coefficient versus supply voltage (V_{DD}) at the TTT corner over the V_{DD} range of 1–2 V.

from $2.09 \mu\text{A}$ at 273°K to $2.71 \mu\text{A}$ at 393°K . Over this range, I_{D5} can be accurately approximated by $0.039 \mu\text{A} \times T^{(0.71)}$. Using $\alpha = 0.71$ and $m = 1.5$, $(\alpha + m - 2)$ is equal to 0.21 . Consequently, Eq. (10) predicts that the magnitude of $|dV_{GS}/dT|$ decreases nonlinearly with increasing temperature. According to Eq. (7), this nonlinear behavior is complementary to the nonlinear component of dV_{BE}/dT , enabling V_{GS} to effectively compensate the curvature of V_{BE} when the two terms are combined.

4. Simulation results

Post-layout simulation results using a 180-nm standard CMOS technology are presented in this section. Fig. 13 shows the active layout area, which occupies $164 \times 148 \mu\text{m}^2$. Fig. 14 shows the simulated V_{ref} at

the TTT corner using the previously introduced digital trimming words a and b , with $a = 1001$ and $b = 01000$ at $V_{DD} = 1.1$ V, resulting in a TC of $0.92 \text{ ppm}/^\circ\text{C}$ and a nominal reference voltage of 599.4 mV .

As discussed in the previous section, V_{DD} dependence can disturb the balance among the PTAT, CTAT1, and CTAT2 components; therefore, the variation of TC with respect to V_{DD} must be carefully evaluated, especially in ultra-low-TC voltage references. Fig. 15 illustrates the TC value versus V_{DD} . As shown, the maximum TC is $1.77 \text{ ppm}/^\circ\text{C}$ over a V_{DD} range of 1 to 2 V. The highest TC occurs at $V_{DD} = 1$ V, where the output resistance of the transistors reaches its minimum value, making the circuit more sensitive to supply variations.

The pseudo-cascode configuration is used to maintain accurate replication of the PTAT and CTAT1 currents in the output branch. Fig. 16(a) and (b) plot the I_{D4b}/I_{D2b} ratio for the pseudo-cascode and conventional current mirrors, respectively, as V_{DD} varies from 1 V to 2 V. In Fig. 16(b), M_{1a} – M_{4a} are omitted to represent the non-cascode case. As shown, the pseudo-cascode current mirror preserves the current ratio significantly more accurately, indicating that the combined PTAT, CTAT1, and CTAT2 components become substantially less sensitive to V_{DD} variations. This allows achieving a low-TC BGR across the entire V_{DD} range.

Fig. 17(a) shows V_{ref} at the SSS corner with $a = 1001$ and $b = 01000$ (the same values used at the TTT corner), resulting in $\text{TC} = 20.9 \text{ ppm}/^\circ\text{C}$ and $V_{ref} = 612.5 \text{ mV}$. After trimming, $V_{ref} = 600.1 \text{ mV}$ and $\text{TC} = 1.02 \text{ ppm}/^\circ\text{C}$ are achieved using $a = 1100$ and $b = 11,100$. Similarly, Fig. 17 (b) presents the results at the FFF corner. With the same initial trimming values ($a = 1001$ and $b = 01000$), $\text{TC} = 18.2 \text{ ppm}/^\circ\text{C}$ and $V_{ref} = 587.9 \text{ mV}$ are obtained. After trimming, $V_{ref} = 600.5 \text{ mV}$ and $\text{TC} = 2.02 \text{ ppm}/^\circ\text{C}$ are achieved using $a = 0100$ and $b = 00000$.

Fig. 18(a) and (b) present the PSRR of the proposed voltage reference at $V_{DD} = 1$ V and 2 V, respectively. The DC PSRR is -81 dB at $V_{DD} = 1$ V and -90.4 dB at $V_{DD} = 2$ V. Due to the parasitic capacitance of the layout, the post-layout PSRR is slightly degraded compared to the pre-layout results. As can be seen, at $V_{DD} = 1$ V and 2 V, the cut off frequencies of the post-layout (pre-layout) simulations are 0.48 KHz (1.3 KHz) and 158 Hz (380 Hz), respectively.

Fig. 19 shows V_{ref} versus supply voltage from 1 V to 2 V. As can be seen, the excellent LS of $0.003\%/V$ is achieved due to the use of the pseudo-cascode configuration.

To evaluate the operation of the startup circuit across different process corners and temperatures, a transient analysis is performed. The supply voltage V_{DD} is ramped from 0 V to 1.1 V within 0.5 s, and the output reference voltage V_{ref} is monitored. Simulations are carried out at the SSS, FFF, and TTT process corners, as well as at 0°C and 120°C . The results are shown in Fig. 20. As can be observed, V_{ref} reaches its final value (approximately 600 mV) after the ramp-up in all cases, with no oscillation or latch-up, indicating that the startup circuit operates correctly under all simulated conditions.

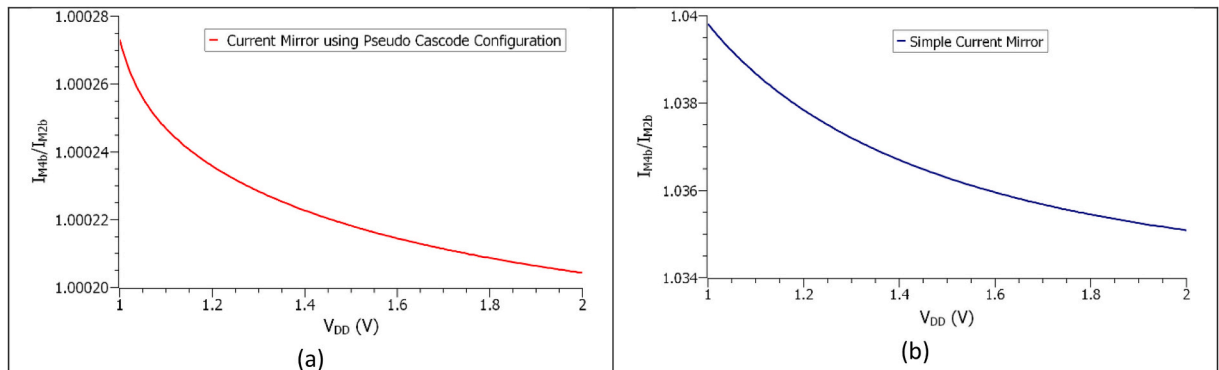


Fig. 16. Current ratio I_{D4b}/I_{D2b} as a function of V_{DD} from 1 V to 2 V. (a) Pseudo-cascode current mirror configuration. (b) Conventional current mirror configuration with M_{1a} – M_{4a} removed.

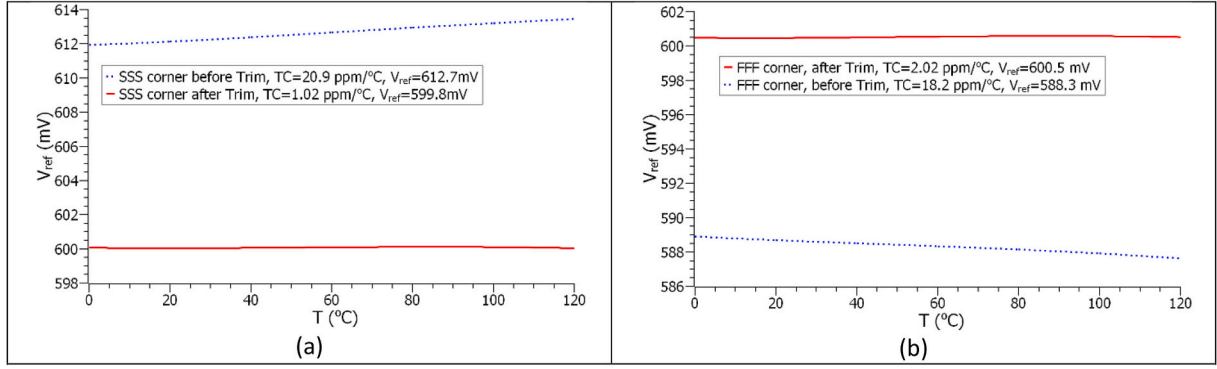


Fig. 17. Simulated reference voltage (V_{ref}) at the SSS and FFF process corners before and after trimming. (a) SSS corner. (b) FFF corner.

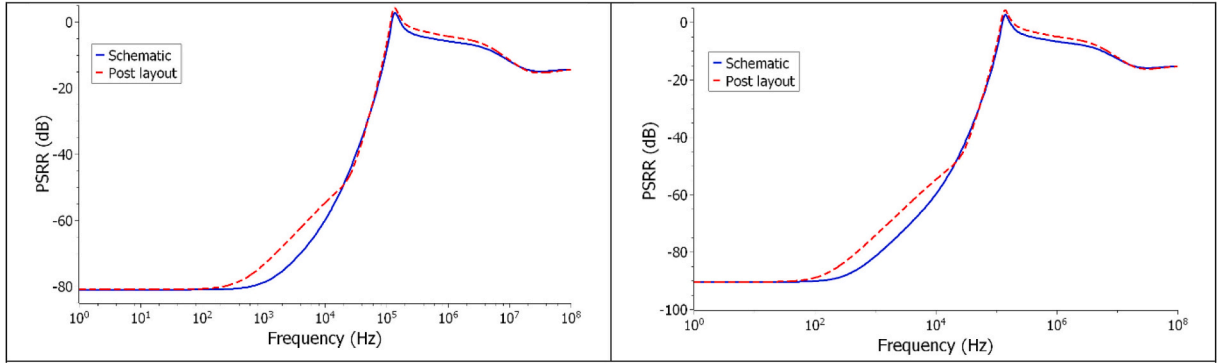


Fig. 18. PSRR of the proposed voltage reference. (a) at $V_{DD} = 1$ V (b) at $V_{DD} = 2$ V.

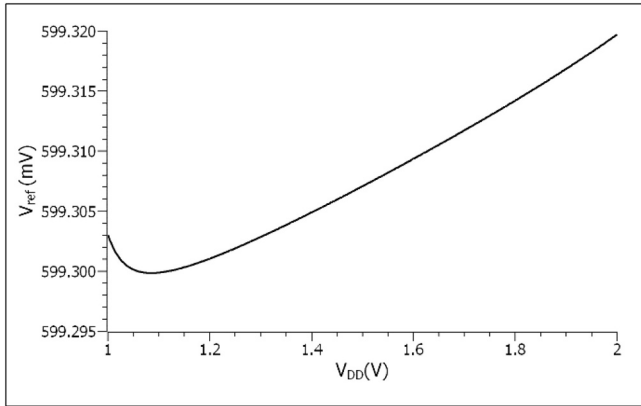


Fig. 19. V_{ref} versus supply voltage from 1 V to 2 V.

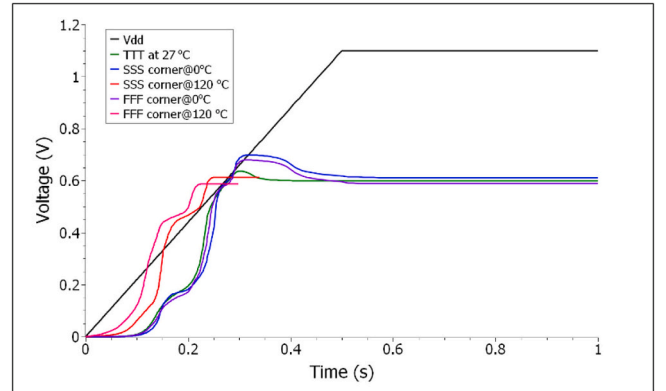


Fig. 20. Startup transient of V_{ref} across process corners and temperatures.

It should be noted that the presented design differs slightly from the pre-layout design (initial schematic). These modifications are introduced to improve the post-layout temperature coefficient and the reference voltage accuracy. In the original pre-layout design, the dimensions of M_5 are $10 \mu\text{m} / 1 \mu\text{m}$; in the current design, they are changed to $9.8 \mu\text{m} / 1 \mu\text{m}$ as listed in Table 1. Furthermore, the value of the main resistor R_4 was $290.1 \text{ k}\Omega$ in the pre-layout design, while in the current layout it is changed to $285.2 \text{ k}\Omega$ as shown in Fig. 10c.

Table 2 lists the key circuit parameters before and after layout. As can be seen, the most significant difference appears in the PSRR value at 100 Hz, which is due to unwanted parasitic capacitances introduced in the layout. Additionally, factors such as the resistance of the interconnect paths, their temperature dependence relative to the main resistors, and the resistance mismatch due to path asymmetries (although layout

Table 2

Trimmed pre/post-layout performance at TTT corner.

Parameters	Pre-layout	Post-layout
TC (ppm/°C)	0.68	0.92
Power (μW)	9.12	9.1
PSRR (dB@1 KHz, $V_{dd} = 1$ V)	-78.8	-73.3
V_{ref} (mV)	600.05	599.4
Line Sensitivity (%/V)	0.003	0.003
PSRR (dB@DC, $V_{dd} = 1$ V)	-80.9	-80.8

symmetry is maintained), together with other parasitic effects, have caused a slight increase in the TC from $0.68 \text{ ppm}/^\circ\text{C}$ to $0.92 \text{ ppm}/^\circ\text{C}$.

Fig. 21(a) and (b) show the Monte Carlo results of V_{ref} over mismatch and process variation for the cases without trimming and after trimming,

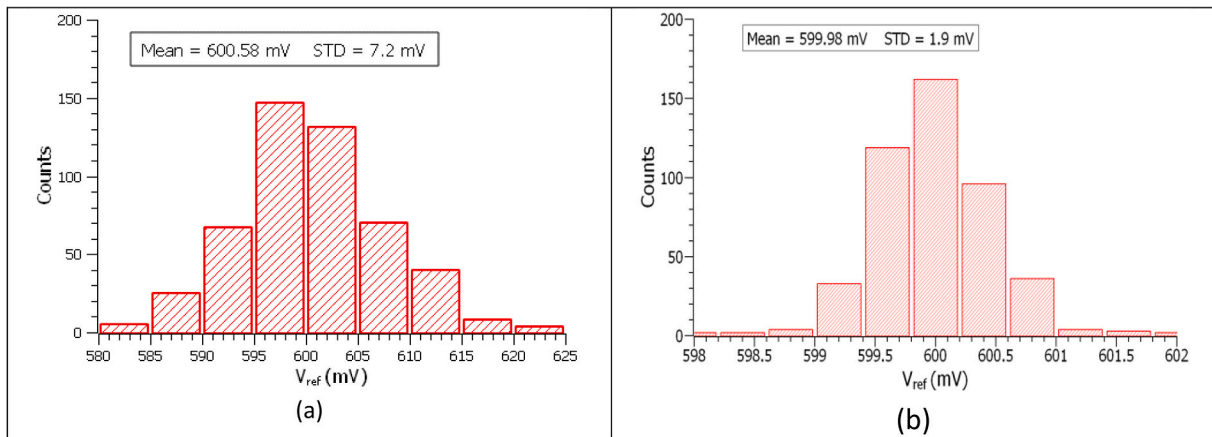


Fig. 21. Monte Carlo results of V_{REF} over 500 runs: (a) without trimming ($a = 1001$, $b = 01000$) and (b) after trimming.

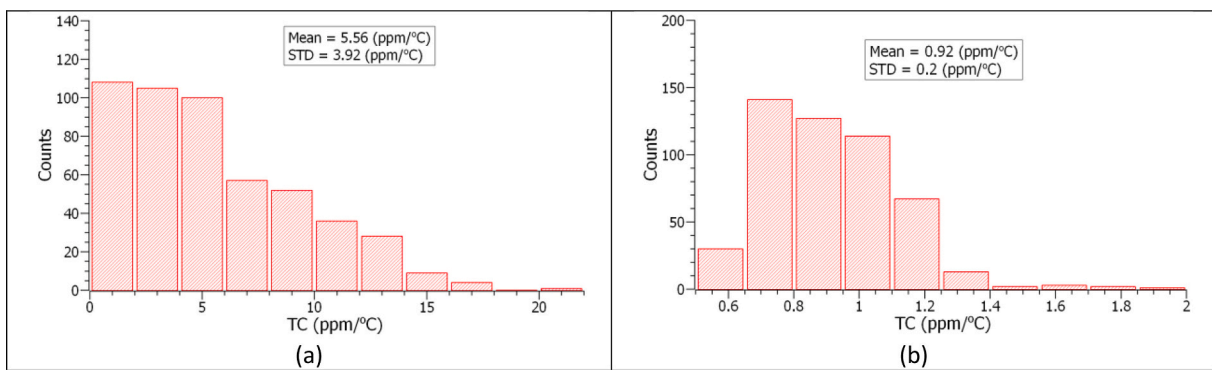


Fig. 22. Monte Carlo results of TC over 500 runs: (a) without trimming and (b) after trimming.

Table 3
Performance Comparison.

	[17]	[16]	[19]	[14]	[15]	This work
Year	2022	2023	2023	2025	2025	2026
Process (nm)	180	65	180	180	180	180
V_{DD} (V)	5	1.8 to 3.4	1.1 to 1.8	3 to 5	1.8	0.9 to 2
Temperature Range (°C)	-55 to 125	-40 to 125	-40 to 125	-40 to 90	-40 to 125	0 to 120
Average TC (ppm/°C)	1.64	0.959	2.4	1.31	16.9	0.92
Power (μ W)	325	138	60	94	0.33	9.1
Line Sensitivity (%/V)	0.2	0.25	0.03	0.006	0.52	0.003
PSRR (dB@DC)	-50	-48	-76	-	-65	-81
PSRR (dB@100 Hz)	-50	-48	-73	-	-65	-80.7
Active Area (mm^2)	0.06	0.037	0.088	0.07	0.027	0.024
Trimming	Yes	Yes	Yes	Yes	Yes	Yes
Sim/Fab	Sim	Sim	Fab	Fab	Sim	Sim
Reference Voltage (V)	1.21	1.2	0.6	0.65	1.24	0.6
FOM_1 (°C ³ /A m ²)	0.5	1.84	0.23	0.68	0.13	7.2
FOM_2 (°C dB/A m ²)	0.08	0.32	0.061	-	0.031	4.03

respectively. Without trimming, using $a = 1001$ and $b = 01000$, the mean value is 600.5 mV with a standard deviation of 7.2 mV. After trimming, the mean is 599.8 mV with a standard deviation of 1.9 mV. In the trimming procedure, for each of the 16 values of digital word a (with fixed b), 500-sample Monte Carlo simulation is performed, and the value that produces V_{ref} closest to 600 mV is selected.

Fig. 22(a) and (b) show the TC distribution for the cases without trimming and after trimming, over mismatch and process variation, respectively. Without trimming, the TC is 5.56 ppm/°C with a standard deviation of 3.9 ppm/°C. For TC trimming, digital word a remains fixed while digital word b is varied. For each of the 32 values of digital word b , a 500-sample Monte Carlo simulation is performed, and the minimum

TC is selected.

Table 3 provides a comparison between the proposed design and other state-of-the-art BJT-based voltage references. Despite its simple structure, the proposed curvature-compensated voltage reference achieves one of the lowest reported TCs among the compared works. Its power consumption is also competitive, except for [15], which consumes less power but exhibits a much higher TC of 16.9 ppm/°C. Moreover, owing to its simplicity, the proposed design attains the smallest active area and the lowest line sensitivity among the compared works. Also, two figures of merit, FOM_1 and FOM_2 , are used to compare the proposed voltage reference with other works. The FOM_s are defined as:

$$FOM_1 = \frac{(T_{\max} - T_{\min})^2}{TC \times I_{\text{supply}} \times \text{Area}} \frac{1}{10^{22}}, FOM_2 = \frac{|PSRR@100\text{Hz}|}{TC \times I_{\text{supply}} \times \text{Area}} \frac{1}{10^{20}} \quad (11)$$

where $(T_{\max} - T_{\min})$ is the temperature range (in °C), TC is the temperature coefficient, I_{supply} is the supply current, Area is the active area and PSRR is in dB.

5. Conclusion

In this work, a curvature-compensated bandgap voltage reference is presented that combines a ΔV_{BE} -based PTAT term with two CTAT components derived from V_{BE} and V_{GS} to effectively suppress the temperature dependence of the output voltage. The proposed architecture not only reduces the TC but also enables independent trimming of both the reference voltage level and the TC. In addition, a pseudo-cascode current mirror enhances robustness against supply voltage variations. Post-layout PVT and Monte Carlo simulation results demonstrate that the circuit generates a reference voltage of approximately 600 mV with an average TC of 0.92 ppm/°C, high PSRR, and a very low LS of about 0.003%/V, while occupying a compact active area.

CRediT authorship contribution statement

Mohammad Rashtian: Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] Kim J, Kim J, Park C, Yang M, Jung W. A wide range, energy-efficient temperature sensor based on direct temperature-voltage comparison. *IEEE Solid-State Circuits Lett* 2023;6:113–6. <https://doi.org/10.1109/LSSC.2023.3266437>.
- [2] Moisello E, Ippolito CM, Bruno G, Malcovati P, Bonizzoni E. A MOS-based temperature sensor with inherent inaccuracy reduction enabled by time-domain operation. *IEEE Trans Instrum Meas* 2023;72:2004110. <https://doi.org/10.1109/TIM.2023.3276016>.
- [3] Adimulam MK, Srinivas MB. A 12-bit, 1.1-GS/s, low-power flash ADC. *IEEE Trans Very Large Scale Integr VLSI Syst* 2022;30(3):277–90. <https://doi.org/10.1109/TVLSI.2021.3138538>.
- [4] Mikulić J, Schatzberger G, Barić A. A 1-MHz on-chip relaxation oscillator with comparator delay cancellation. In: *ESSCIRC 2017 - 43rd IEEE European Solid State Circuits Conference*, Leuven, Belgium; 2017. p. 95–8. <https://doi.org/10.1109/ESSCIRC.2017.8094534>.
- [5] Zhang H, et al. A nano-watt MOS-only voltage reference with high-slope PTAT voltage generators. *IEEE Trans Circuits Syst II Express Briefs* 2018;65(1):1–5. <https://doi.org/10.1109/TCSII.2017.2654441>.
- [6] Rashtian M. MOS only nano-watt sub-bandgap voltage reference. *Analog Integr Circ Sig Process* 2023;114(3):371–7. <https://doi.org/10.1007/s10470-023-02133-3>.
- [7] Li Y, Caginalp R, Lee I. A picowatt CMOS voltage reference using independent TC and output level calibrations. *IEEE Trans Very Large Scale Integr VLSI Syst* 2025;33(5):1244–54. <https://doi.org/10.1109/TVLSI.2024.3508259>.
- [8] Lin J, Wang L, Zhan C, Lu Y. A 1-nW ultra-low voltage subthreshold CMOS voltage reference with 0.0154%/V line sensitivity. *IEEE trans circuits Syst II express. Briefs* 2019;66(10):1653–7. <https://doi.org/10.1109/TCSII.2019.2920693>.
- [9] Rashidian H, Soltani I, Maghsoudi M. A 75.12-nW 0.5-V MOS-based BGR comprising a curvature compensation circuit for analog-to-digital converter applications. *AEU Int J Electron Commun* 2025;191:155678. <https://doi.org/10.1016/j.aeue.2025.155678>.
- [10] Olivera F, Petraglia A. Adjustable output CMOS voltage reference design. In: *IEEE international symposium on circuits and systems (ISCAS)*, Seville, Spain. 2020; 2020. p. 1. <https://doi.org/10.1109/ISCAS45731.2020.9181000>.
- [11] Ji J, Li Y, Zhao Y. A 5.56 ppm/°C curvature-compensated CMOS voltage reference using reverse body biasing technique. *AEU Int J Electron Commun* 2025;202:155995. <https://doi.org/10.1016/j.aeue.2025.155995>.
- [12] Liu X, Liang S, Liu W, Sun P. A 2.5 ppm/°C voltage reference combining traditional BGR and ZTC MOSFET high-order curvature compensation. *IEEE trans circuits Syst II express. Briefs* 2021;68(4):1093–7. <https://doi.org/10.1109/TCSII.2020.3027768>.
- [13] Banba H, Shiga H, Umezawa A, Miyaba T, Tanzawa T, Atsumi S, et al. A CMOS bandgap reference circuit with sub-1-V operation. *IEEE J Solid State Circuits* 1999; 34(5):670–4. <https://doi.org/10.1109/4.760378>.
- [14] Lv W, Cai Y, Wang Y, Chen Z. A 1.31-ppm/°C CMOS voltage reference with second-order and sub-ranging compensation. *IEEE Trans Circuits Syst I Regul Pap* 2025;72(11):6378–88. <https://doi.org/10.1109/TCSI.2025.3562148>.
- [15] Yan Z, Zhang B, Yang R, Zheng Y, Li J, Luo Z, et al. A 1.2-V compact bandgap reference with curvature compensation technology. *IEICE Electron Express* 2025; 22:20250028. <https://doi.org/10.1587/ele.22.20250028>.
- [16] Chen Q, Gong Y, Tu R, Xie J, Dai H, Chen Y, et al. A 1.2-V 0.959-ppm/°C multi-section curvature-compensated bandgap voltage reference with trimming. *Microelectron J* 2023;136:105769. <https://doi.org/10.1016/j.mejo.2023.105769>.
- [17] Li T, Gao B, Zhao X, Gong M. A curvature-compensated bandgap voltage reference with a temperature coefficient trimming circuit. *Microelectron J* 2022;119:105308. <https://doi.org/10.1016/j.mejo.2021.105308>.
- [18] Huang S, Li M, Li H, et al. A sub-1 ppm/°C bandgap voltage reference with high-order temperature compensation in 0.18-μm CMOS process. *IEEE Trans Circuits Syst I Regul Pap* 2022;69(4):1408–16. <https://doi.org/10.1109/TCSI.2021.3139908>.
- [19] Liao X, Zhang Y, Zhang S, Liu L. A 3.0 μVrms, 2.4 ppm/°C BGR with feedback coefficient enhancement and bowl-shaped curvature compensation. *IEEE Trans Circuits Syst I Regul Pap* 2024;71(5):2424–33. <https://doi.org/10.1109/TCSI.2024.3373788>.
- [20] Razavi B. *Design of analog CMOS integrated circuits*. 2nd ed. New York: McGraw-Hill Education; 2017.
- [21] Duan Q, Lin C, Liu P, Huang S, Meng Z. A 2.1 ppm/°C, 0.55–2.4 V, 5.6 nW, 235 mV, CMOS-only subthreshold voltage reference. *Microelectron J* 2021;113:105060. <https://doi.org/10.1016/j.mejo.2021.105060>.
- [22] Malcovati P, Maloberti F, Flocchi C, Pruzzi M. Curvature-compensated BiCMOS bandgap with 1-V supply voltage. *IEEE J Solid State Circuits* 2001;36(7):1076–81. <https://doi.org/10.1109/4.933463>.